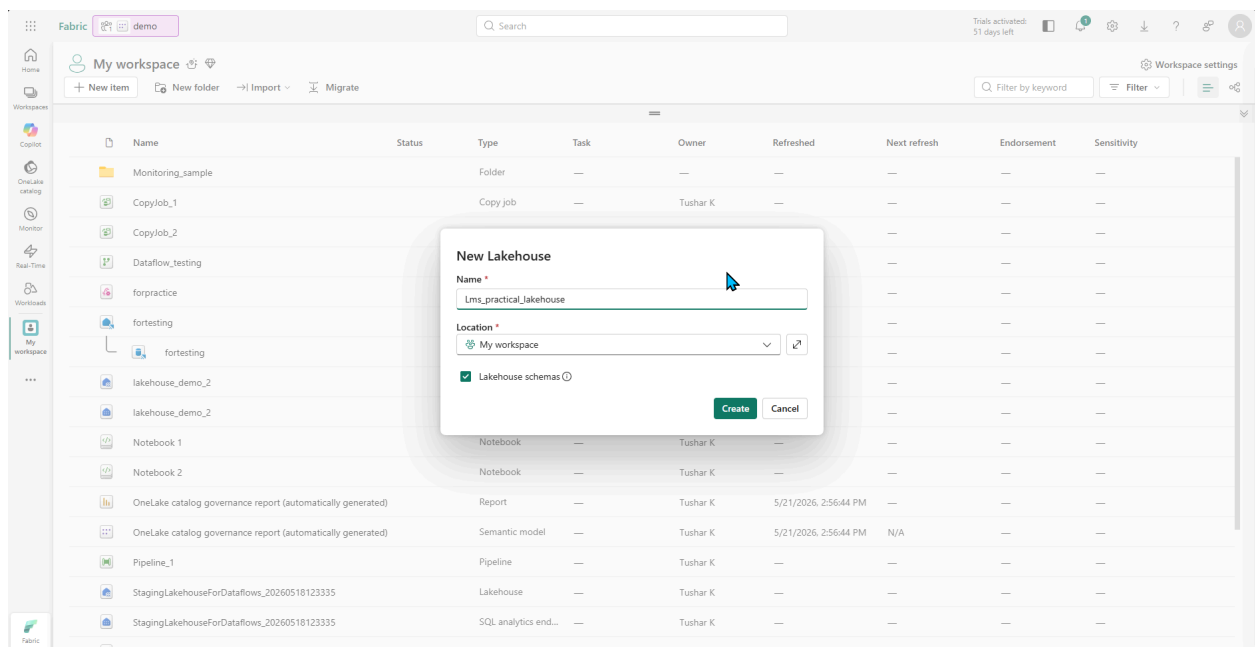
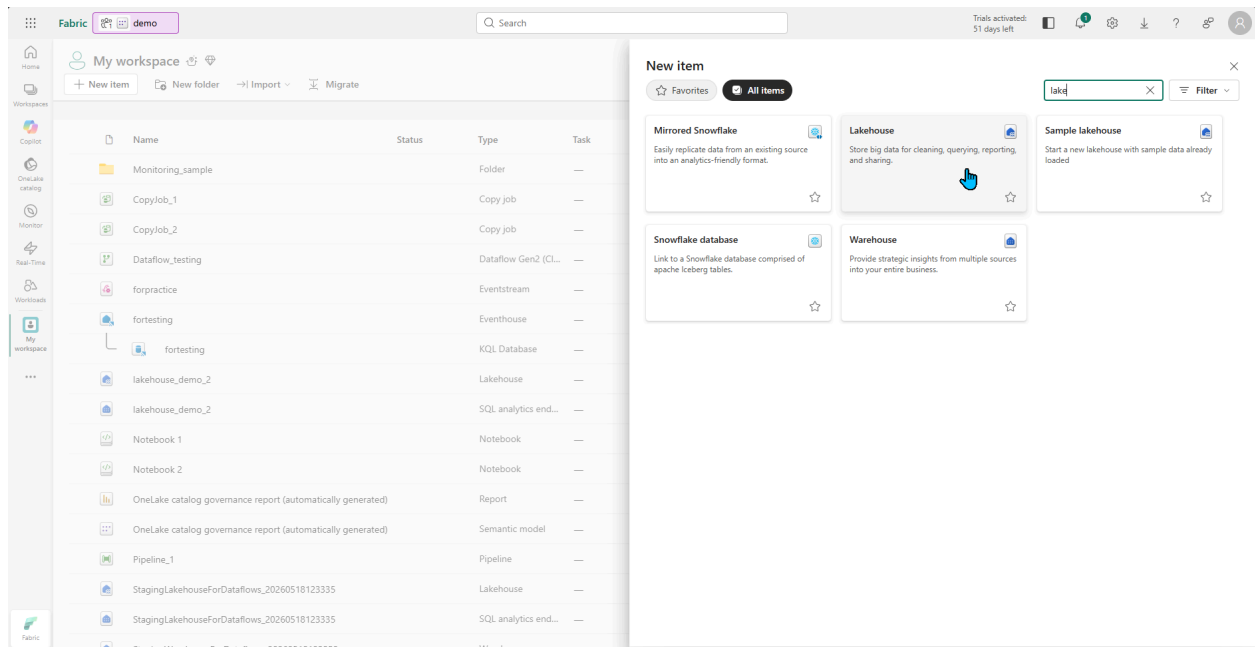


Fabric assignment

Task 1: OneLake & Lakehouse

- Create a Lakehouse in Microsoft Fabric
- Upload the provided CSV file to the Files section
- Load the data into Lakehouse Tables
- Verify the table using the SQL Endpoint



Fabric demo Lms_practical_lakehouse Search

Home Materialized lake views

Get data Get data New semantic model Open notebook Add to data agent Manage OneLake security Update all variables

For optimal refresh to work efficiently, enable the delta CDF property on both source tables and materialized lake views used as source.

A SQL analytics endpoint for SQL querying was created with this item.

Explorer

Add lakehouses

Lms_practical_lakehouse

Tables

dbo

Files

Get data in your lakehouse

Upload files

Upload data from your local machine.

Start with sample data

Automatically import tables filled with sample data.

New shortcut

Access data that resides in an external lake.

New Data

Prep, clear, ingest data

Upload files

My workspace/Lms_practical_lakehouse/Files/

Overwrite if files already exist

Upload

Current uploads

File Name	Lakehouse Name	Dismiss	Completed	All
user_activity.csv	Lms_practical_lakehouse	901 B / 901 B		
activity_logs.csv	Lms_practical_lakehouse	611 B / 611 B		

Fabric Lms_practical_lakehouse Notebook 3 Search

Home Materialized lake views

Get data Get data New semantic model Open notebook Add to data agent Manage OneLake security Update all variables

For optimal refresh to work efficiently, enable the delta CDF property on both source tables and materialized lake views used as source.

A SQL analytics endpoint for SQL querying was created with this item.

Explorer

Add lakehouses

Lms_practical_lakehouse

Tables

dbo

Files

My workspace > Lms_practical_lakehouse > Files

Showing 2 items

Search files

Name	Date modified	Type	Size
activity_logs.csv	6/25/2024 11:28:33 AM	csv	611 B
user_activity.csv		csv	901 B

Open in new tab

Copy URL

Load to Tables

New table

Existing table

View

Download

Rename

Delete

Properties

Succeeded (0 sec 84 ms)

Folders 0 Files 2

Fabric Lms_practical_lakehouse Notebook 3

Home Materialized lake views

Get data New semantic model Open notebook Add to data agent Manage OneLake security Update all variables

For optimal refresh to work efficiently, enable the delta CDF property on both source tables and materialized lake views as source.

A SQL analytics endpoint for SQL querying was created with this item.

Explorer

My workspace > Lms_practical_lakehouse > Files

Showing 2 items

activity_logs.csv 611 B

user_activity.csv 901 B

Load file to new table

All fields marked with * are required

Schema * dbo

New table name * user_activity

Column header ☒ Use header for column names

Separator ,

Separator cannot use the following characters: (){}|' "

Load Cancel

Succeeded (0 sec 84 ms)

Folders 0 Files 2

Fabric Lms_practical_lakehouse Lms_practical_lakehouse

Home Security Management Help

New SQL query Monitor New semantic model New API for GraphQL Download SQL database project Open in Copilot

Explorer

SQL query 1

```
1 SELECT TOP (100) [Timestamp],
2 [UserId],
3 [EventType],
4 [Status]
5 FROM [Lms_practical_lakehouse].[dbo].[activity_logs]
```

Messages Results

	Timestamp	UserId	EventType	Status
1	2025-01-01 09:05:00.000000	U101	Login	Success
2	2025-01-01 09:10:00.000000	U102	Login	Success
3	2025-01-01 10:00:00.000000	U103	Login	Failed
4	2025-01-01 10:15:00.000000	U101	FileUpload	Success
5	2025-01-01 10:30:00.000000	U102	FileDownload	Success
6	2025-01-01 09:00:00.000000	U105	Login	Success
7	2025-01-02 09:00:00.000000	U101	Login	Success
8	2025-01-02 10:30:00.000000	U104	FileUpload	Failed
9	2025-01-02 11:30:00.000000	U106	Login	Success
10	2025-01-03 09:15:00.000000	U103	Login	Success
11	2025-01-03 10:00:00.000000	U106	FileUpload	Success
12	2025-01-03 10:30:00.000000	U101	Logout	Success

Succeeded (3 sec 110 ms) Copilot completions: Off

Columns: 4 Rows: 15

The screenshot shows the Microsoft Fabric SQL query editor. The query is as follows:

```
SELECT TOP (100) [UserId],
[UserName],
[ActivityType],
[ActivityDate],
[ActivityTime],
[Country]
FROM [Lms_practical_lakehouse].[dbo].[user_activity]
```

The results table displays 12 rows of data:

ARI	UserId	ARI	UserName	ARI	ActivityType	ActivityDate	ActivityTime	ARI	Country
1	U101		Rahul		Login	2025-01-01	2026-05-22 09:05:00.000000		India
2	U102		Anita		Login	2025-01-01	2026-05-22 09:10:00.000000		India
3	U103		John		Login	2025-01-01	2026-05-22 10:00:00.000000		USA
4	U101		Rahul		FileUpload	2025-01-01	2026-05-22 10:15:00.000000		India
5	U102		Anita		FileDownload	2025-01-01	2026-05-22 10:30:00.000000		India
6	U104		Emma		Login	2025-01-01	2026-05-22 11:00:00.000000		UK
7	U103		John		Logout	2025-01-01	2026-05-22 11:30:00.000000		USA
8	U105		Arjun		Login	2025-01-02	2026-05-22 09:00:00.000000		India
9	U101		Rahul		Login	2025-01-02	2026-05-22 09:05:00.000000		India
10	U104		Emma		FileUpload	2025-01-02	2026-05-22 10:20:00.000000		UK
11	U102		Anita		Logout	2025-01-02	2026-05-22 10:45:00.000000		India
12	U105		Arjun		FileDownload	2025-01-02	2026-05-22 11:10:00.000000		India

Task 2: Dataflow Gen2

- Create a Dataflow Gen2
- Use Lakehouse table or CSV as the source
- Perform basic transformations:
 - a) Convert activity date to Date format
 - b) Derive ActivityHour from activity time
 - c) Rename columns for clarity
- Load the transformed data back to the same Lakehouse as a new table.

The screenshot shows the Microsoft Fabric workspace. The 'New item' dialog is open, displaying various options for creating new items. The 'Dataflow Gen2' option is highlighted, indicating the next step in the task.

Fabric | lms_practical_dataflowgen2 | Search

Power Query | Draft saved | Search (Alt + Q)

Get data

Choose data

Search

Display options

- lms_practical_lakehouse [15]
- Files
 - queryinsights_exec_requests...
 - queryinsights_exec_sessions...
 - queryinsights_frequently_run...
 - queryinsights_long_running...
 - queryinsights_sql_job_inights...
 - sysdm_db_external_tables_lo...
 - sysexternal_delta_tables
 - sysmanaged_delta_table_che...
 - sysmanaged_delta_table_foris...
 - sysmanaged_delta_table_log...
 - sysmanaged_delta_tables
 - sysstm_db_schemas
 - activity_logs
 - user_activity**

Userid	Username	ActivityType	ActivityDate	ActivityTime	Country
U101	Rahul	Login	1/1/2025	5/22/2026 9:05:00 AM	India
U102	Anita	Login	1/1/2025	5/22/2026 9:10:00 AM	India
U103	John	Login	1/1/2025	5/22/2026 10:00:00 AM	USA
U101	Rahul	FileUpload	1/1/2025	5/22/2026 10:15:00 AM	India
U102	Anita	FileDownload	1/1/2025	5/22/2026 10:30:00 AM	India
U104	Emma	Login	1/1/2025	5/22/2026 11:00:00 AM	UK
U103	John	Logout	1/1/2025	5/22/2026 11:30:00 AM	USA
U105	Ajgun	Login	1/2/2025	5/22/2026 9:00:00 AM	India
U101	Rahul	Login	1/2/2025	5/22/2026 9:05:00 AM	India
U104	Emma	FileUpload	1/2/2025	5/22/2026 10:20:00 AM	UK
U102	Anita	Logout	1/2/2025	5/22/2026 10:45:00 AM	India
U105	Ajgun	FileDownload	1/2/2025	5/22/2026 11:10:00 AM	India
U106	Sophia	Login	1/2/2025	5/22/2026 11:30:00 AM	Canada
U103	John	Login	1/3/2025	5/22/2026 9:15:00 AM	USA
U106	Sophia	FileUpload	1/3/2025	5/22/2026 10:00:00 AM	Canada
U101	Rahul	Logout	1/3/2025	5/22/2026 10:30:00 AM	India
U104	Emma	Login	1/3/2025	5/22/2026 11:00:00 AM	UK
U105	Ajgun	Logout	1/3/2025	5/22/2026 11:20:00 AM	India
U102	Anita	Login	1/3/2025	5/22/2026 11:45:00 AM	India
U106	Sophia	Logout	1/3/2025	5/22/2026 12:00:00 PM	Canada

Back Cancel Create

Fabric | lms_practical_dataflowgen2 | Search

Power Query | Draft saved | Search (Alt + Q)

Home

Transform Add column View Help

Queries [2]

- activity_logs
- user_activity**

Update method

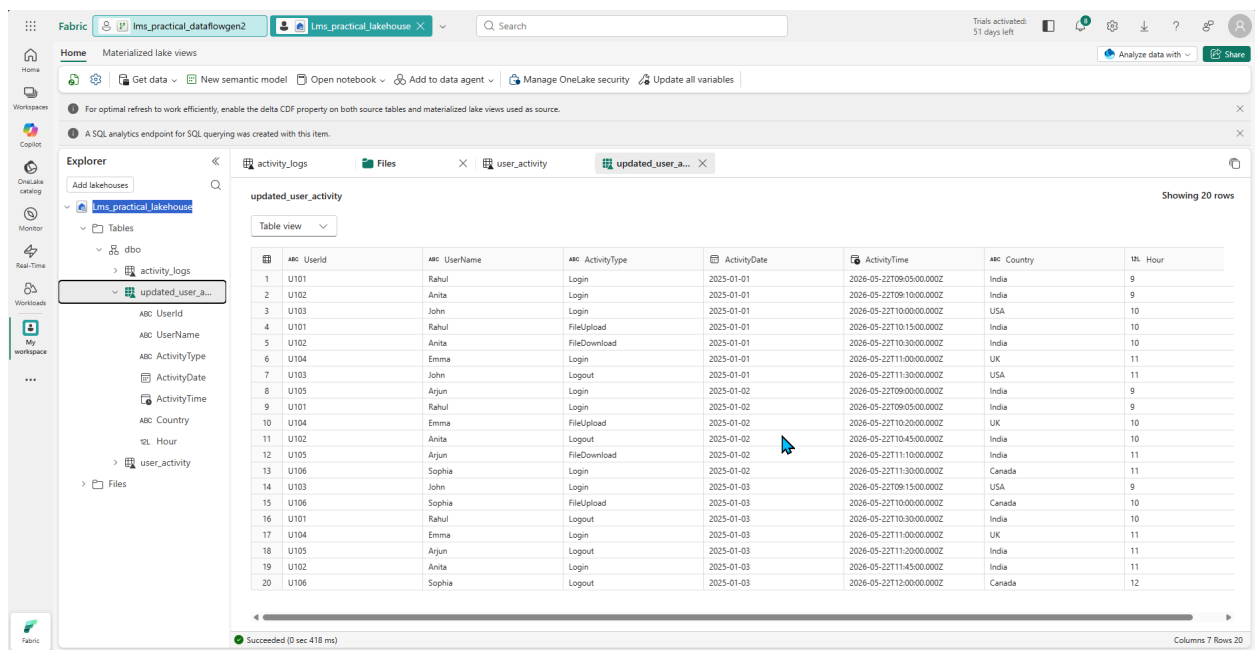
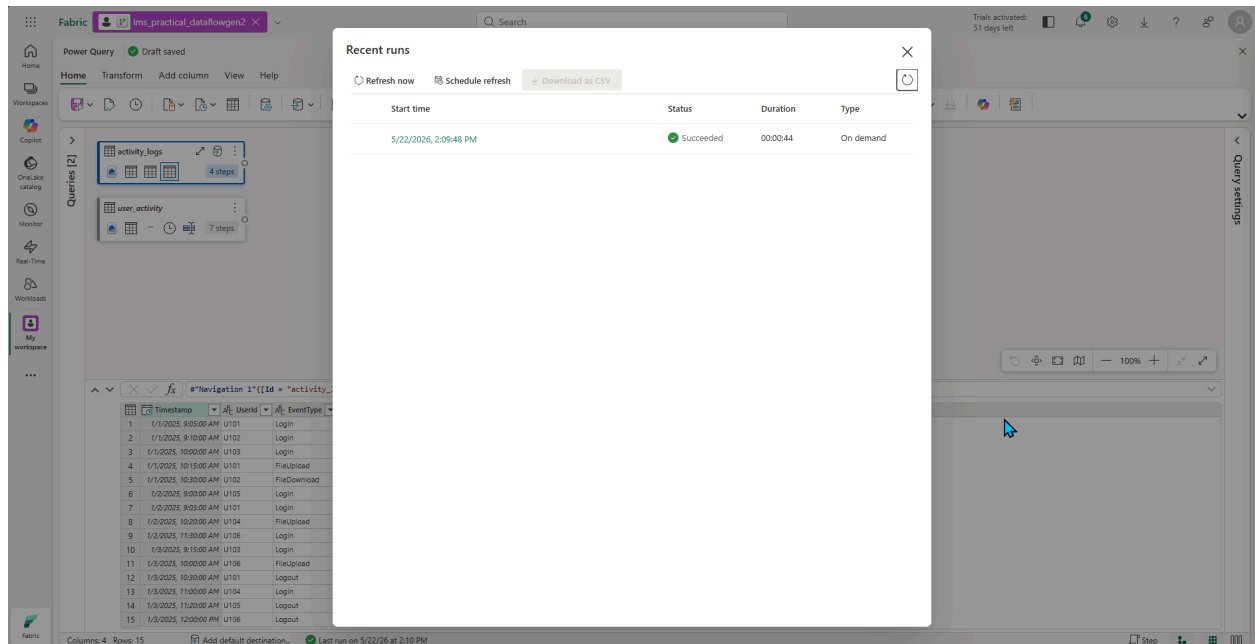
Replace

Column mapping

Auto mapping and allow for schema change

Type	ActivityDate	ActivityTime	Country	Hour
1	1/1/2025	5/22/2026 9:05:00 AM	India	9
2	1/1/2025	5/22/2026 9:10:00 AM	India	9
3	U103	John	Login	10
4	U101	Rahul	FileUpload	10
5	U102	Anita	FileDownload	10
6	U104	Emma	Login	11
7	U103	John	Logout	11
8	U105	Ajgun	Login	9
9	U101	Rahul	Login	9
10	U104	Emma	FileUpload	10
11	U102	Anita	Logout	10
12	U105	Ajgun	FileDownload	11
13	U106	Sophia	Login	11
14	U103	John	Login	9
15	U106	Sophia	FileUpload	10

Columns: 7 Rows: 20 Add default destination... Not run Step



Task 3: KQL Analytics

- Create a KQL Database in Fabric
- Load sample or generated log data
- Write KQL queries to analyze:
 - a) Total event count
 - b) Events by event type
 - c) Successful vs failed logins

Fabric | lms_practical_dataflowgen2 | Lms_practical_lakehouse | LMS_KQL | LMS_KQL | Search | Trials activated: 51 days left

Eventhouse | LMS_KQL - My workspace

Live view | Database | Always-on | Plugins | Details

LMS_KQL

- System overview
- Databases
- Monitoring

KQL databases

- LMS_KQL

System overview

Eventhouse storage

Original size: 0 B
Compressed size: 0 B

Storage resources

Eventhouse storage (compressed): 0 B

OneLake availability

Activity in minutes

1 Minutes
1 Databases

Ingestion

Ingestion data is not available

Top 10 queried databases | Top 10 ingested databases

Partial ingestion data To get additional data turn on fabric workspace monitoring.

Workspace settings

Name	Queries	Errors	Duration	Cache misses
------	---------	--------	----------	--------------

Eventhouse Details

Region: Central India
Query URI: Copy URI
Last ingestion: N/A
Ingestion URI: Copy URI
Always-on: Off
Python plugin: Not installed

Related elements

- LMS_KQL
- LMS_KQL

Fabric | lms_practical_dataflowgen2 | Lms_practical_lakehouse | LMS_KQL | LMS_KQL | Search | Trials activated: 51 days left

Source | Configure | Inspect | Summary

Get data Summary

OneLake → LMS_KQL/analitics-data

Data Preparation Details

- Create table (analitics-data) -
- Create mapping (analitics-data_mapping) -
- Data queuing -

What can you do with the data?

Now that you've ingested data, you can explore, run queries, and visualize the data using the dashboard

Explore the results

Search, sort, filter, and expand rows to highlight trends or problems.

Explore

Undo ingestion

Delete the data you've just ingested.

Delete ingested data

Create new dashboard

View your ingested data's key metrics, sample records, and ingestion history

Create dashboard

Close

Fabric | lms_practical_dataflowgen2 | Lms_practical_lakehouse | LMS_KQL | LMS_KQL X

Eventhouse Database Table

Explore data | Get data | Query with code | Real-Time Dashboard | KQL Queryset | Power BI report | Anomaly Detector | Notebook | View files | OneLake | Rename | Edit schema | Show ingestion mappings | Delete table

Workspaces

System overview

Databases

Monitoring

KQL databases

LMS_KQL

LMS_KQL_queryset

Tables

analytics-data

Ingestion Mappings

Update Policy

Timestamp

Userid

EventType

Status

Shortcuts

Materialized views

Functions

Database / Tables / analytics-data

Data Activity Tracker

Ingestion: 15 rows | Last run: 2:49:29 PM

1H 6H 1D 3D 7D 30D Interval: 12 hours

May 19, 12:00 AM May 19, 12:00 PM May 20, 12:00 AM May 20, 12:00 PM May 21, 12:00 AM May 21, 12:00 PM May 22, 12:00 AM

Data preview Schema insights

Search keyword

Timestamp	Userid	EventType	Status	IngestionTime
2025-01-01T09:05:00.000Z	U101	Login	Success	2026-05-22T09:18:56.3352929Z
2025-01-01T09:10:00.000Z	U102	Login	Success	2026-05-22T09:18:56.3352929Z
2025-01-01T10:00:00.000Z	U103	Login	Failed	2026-05-22T09:18:56.3352929Z
2025-01-01T10:15:00.000Z	U101	FileUpload	Success	2026-05-22T09:18:56.3352929Z
2025-01-01T10:30:00.000Z	U102	FileDownload	Success	2026-05-22T09:18:56.3352929Z
2025-01-02T09:00:00.000Z	U105	Login	Success	2026-05-22T09:18:56.3352929Z
2025-01-02T09:05:00.000Z	U101	Login	Success	2026-05-22T09:18:56.3352929Z
2025-01-02T10:30:00.000Z	U104	FileUpload	Failed	2026-05-22T09:18:56.3352929Z
2025-01-02T11:30:00.000Z	U106	Login	Success	2026-05-22T09:18:56.3352929Z
2025-01-03T08:15:00.000Z	U103	Login	Success	2026-05-22T09:18:56.3352929Z
2025-01-03T10:00:00.000Z	U108	FileUpload	Success	2026-05-22T09:18:56.3352929Z
2025-01-03T10:30:00.000Z	U101	Logout	Success	2026-05-22T09:18:56.3352929Z
2025-01-03T11:00:00.000Z	U104	Login	Success	2026-05-22T09:18:56.3352929Z
2025-01-03T11:20:00.000Z	U105	Logout	Success	2026-05-22T09:18:56.3352929Z
2025-01-03T12:00:00.000Z	U106	Logout	Success	2026-05-22T09:18:56.3352929Z

Top 15 records from the last 365 Days, limited by the hot cache retention

Table details

Compressed Original

2.1KB 803B

OneLake

Availability Disabled

When enabled, all data in this table will be available in OneLake. [Learn more](#)

Overview

Row count 15

Rows ingested last 24h 15

Schema last altered by fabric@tusharkoit...

Last ingestion May 22, 2026

Ingestion URI Copy URI

Caching policy Unlimited

Retention policy Unlimited

Related elements

Find by name

analytics-data_mapping (csv)

Fabric | lms_practical_dataflowgen2 | Lms_practical_lakehouse | LMS_KQL | KustoQueryWorkbench 1 X

Home Help

Save | Add data source | Copilot

Tab

Explorer

Search

LMS_KQL

Tables

analytics-data

EventType

Status

Timestamp

Userid

Materialized Views

Shortcuts

Entity Groups

Functions

Data sources

LMS_KQL

Run Preview Recall Share query Save to Dashboard KQL Tools Export to CSV Power BI report Add alert

```
1 ["analytics-data"]
2 | count
3
```

Table 1 + Add visual Stats

Search 2026-05-22 09:36 (UTC) Done (0.166 s) # 1 records Hide empty columns

Count
15

The screenshot shows the Microsoft Fabric Dataflow Gen2 interface. The top navigation bar includes tabs for 'lms_practical_dataflowgen2', 'lms_practical_lakehouse', 'LMS_KQL', and 'KustoQueryWorkbench_1'. The left sidebar contains navigation options like Home, Help, Save, Add data source, and Copilot. The main workspace displays a query in the 'Query Editor' tab:

```
1 ["analytics-data"]
2 | summarize TotalEvents = count() by EventType
```

Below the query editor, the 'Table 1' results are shown:

EventType	TotalEvents
Login	8
FileUpload	3
FileDownload	1
Logout	3

The screenshot shows the Microsoft Fabric Dataflow Gen2 interface with a filtered query. The query in the 'Query Editor' tab is:

```
1 ["analytics-data"]
2 | where EventType == "Login"
3 | summarize TotalLogins = count() by Status
```

Below the query editor, the 'Table 1' results are shown:

Status	TotalLogins
Success	7
Failed	1

Task 4: Power BI Report (Fabric)

Create a one-page Power BI report inside Fabric using Dataflow Gen2 output with the following visuals:

- **Card:** Total Users
- **Bar Chart:** Activity count by Activity Type
- **Line Chart:** Activity trend by date
- **Table:** UserName, ActivityType, ActivityDate, Country
- **Slicer:** Country or ActivityType

